STATISTICS IS THE GRAMMAR OF SCIENCE

PROBABILITY AND STATISTICS

LECTURE – 20

LIMIT THEOREMS

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SOME IMPORTANT INEQUALITIES

There are several inequalities used in limit theorems, we shall discuss few of them including'

- Markov's Inequality
- Chebyshev's Inequality

MARKOV'S INEQUALITY

Let X *be a nonnegative random variable; then for any value* $a > 0$

$$P(X \geq a) \leq \frac{E(X)}{a}$$

Proof. For $a > 0$, let

$$I = \begin{cases} 1 & \text{if } X \geq a \\ 0 & \text{otherwise} \end{cases}$$

and note that, since $X \geq 0$,

$$I \leq \frac{X}{a}$$

Taking expectations of the preceding inequality yields

$$E[I] \leq \frac{E[X]}{a}$$

which, because $E[I] = P\{X \geq a\}$, proves the result. □

EXAMPLE A post office, on average, handles 10,000 letters per day. What can be said about the probability that it will handle
(a) at least 15,000 letters tomorrow?
(b) fewer than 15,000 letters tomorrow?

Solution: Let X be the number of letters that this post office will handle tomorrow. Then $E(X) = 10,000$.

(a) By Markov's inequality,

$$P(X \geq 15,000) \leq \frac{E(X)}{15,000} = \frac{10,000}{15,000} = \frac{2}{3}.$$

(b) Using the inequality obtained in (a), we have

$$P(X < 15,000) = 1 - P(X \geq 15,000) \geq 1 - \frac{2}{3} = \frac{1}{3}. \quad \blacklozenge$$

CHEBYSHEV'S INEQUALITY

Let X be a random variable with finite mean μ *and variance* σ^2 *, then, for any value* $k > 0$,

$$P\left(|X-\mu|<k\sigma\right)\geq 1-\frac{1}{k^2}$$

In Simple words we can say that

The probability that any random variable X will assume a value within 'k' standard deviations of the mean is at least $1-\frac{1}{k^2}$.

$$i.e.\ P\left(\mu-k\sigma<X<\mu+k\sigma\right)\geq 1-\frac{1}{k^2}$$

Proof. Since $(X-\mu)^2$ is a nonnegative random variable, we can apply Markov's inequality (with $a=k^2$) to obtain

$$P\{(X-\mu)^2\geq k^2\}\leq\frac{E[(X-\mu)^2]}{k^2} \qquad (2.1)$$

But since $(X-\mu)^2\geq k^2$ if and only if $|X-\mu|\geq k$, Equation (2.1) is equivalent to

$$P\{|X-\mu|\geq k\}\leq\frac{E[(X-\mu)^2]}{k^2}=\frac{\sigma^2}{k^2}$$

and the proof is complete. □

EXAMPLE-1 A random variable X has a mean $\mu=8$ and variance $\sigma^2=9$ and an unknown probability distribution. Find

$(a)\ P(-4<X<20)$

$(b)\ P(|X-8|<6)$

SOLUTION *Here* $\mu=8$ & $\sigma^2=9$. We know that Chebyshev's Inequality is given by

$$P\left(\mu-k\sigma<X<\mu+k\sigma\right)\geq 1-\frac{1}{k^2}$$

$(a)\ P(-4<X<20)$

$\Rightarrow \mu-k\sigma=-4$ *and* $\mu+k\sigma=20$

subtracting two equations we get $2k\sigma=24\Rightarrow 2k=\frac{24}{\sigma}=\frac{24}{3}=8\Rightarrow k=4$

So $P(-4<X<20)\geq 1-\frac{1}{k^2}=1-\frac{1}{4^2}=1-\frac{1}{16}=\frac{15}{16}$ *So* $P(-4<X<20)\geq\frac{15}{16}$

$(b)\ P(|X-8|<6)$

$= P(-6<X-8<6)$

$= P(2<X<14)$

$\Rightarrow \mu - k\sigma = 2 \quad and \quad \mu + k\sigma = 14$

subtracting two equations we get $2k\sigma = 12 \Rightarrow 2k = \frac{12}{\sigma} = \frac{12}{3} = 4 \Rightarrow k = 2$

So $P(2<X<14) \geq 1 - \frac{1}{k^2} = 1 - \frac{1}{2^2} = 1 - \frac{1}{4} = \frac{3}{4}$, *So* $P(2<X<14) \geq \frac{3}{4}$

EXAMPLE-2 A random variable X has a mean $\mu = 10$ and variance $\sigma^2 = 4$ and an unknown probability distribution. Find

$(a)\ P(5<X<15)$

$(b)\ P(|X-10| \geq 3)$

(c) *the value of cons*$\tan t$ *c such that* $P(|X-10| \geq c) \leq 0.04.$

SOLUTION *Here* $\mu = 10$ & $\sigma^2 = 4$. We know that Chebyshev's Inequality is given by

$$P(\mu - k\sigma < X < \mu + k\sigma) \geq 1 - \frac{1}{k^2}$$

$(a)\ P(5<X<15)$

$\Rightarrow \mu - k\sigma = 5 \quad and \quad \mu + k\sigma = 15$

subtracting two equations we get $2k\sigma = 10 \Rightarrow 2k = \frac{10}{2} = 5 \Rightarrow k = \frac{5}{2} = 2.5$

So $P(5<X<15) \geq 1 - \frac{1}{(5/2)^2} = 1 - \frac{1}{25/4} = 1 - \frac{4}{25} = \frac{21}{25}$ *So* $P(5<X<15) \geq \frac{21}{25}$

$(b)\ P(|X-10| \geq 3)$

$= 1 - P(|X-10| < 3)$

$= 1 - P(-3<X-10<3)$

$= 1 - P(7<X<13) ----(1)$

Lets evaluate $P(7<X<13)$

$\Rightarrow \mu - k\sigma = 7 \quad and \quad \mu + k\sigma = 13$

subtracting two equations we get $2k\sigma = 6 \Rightarrow 2k = \frac{6}{\sigma} = \frac{6}{2} = 3 \Rightarrow k = \frac{3}{2}$

So $P(7<X<13) \geq 1 - \frac{1}{(3/2)^2} = 1 - \frac{1}{9/4} = 1 - \frac{4}{9} = \frac{5}{9}$ *So* $P(7<X<13) \geq \frac{5}{9}$

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$Now\, P(7 < X < 13) \geq \frac{5}{9}$

$\Rightarrow P(|X - 10| < 3) \geq \frac{5}{9}$

$\Rightarrow -P(|X - 10| < 3) \leq -\frac{5}{9}$

$\Rightarrow 1 - P(|X - 10| < 3) \leq 1 - \frac{5}{9} = \frac{4}{9}$

$\Rightarrow P(|X - 10| \geq 3) \leq \frac{4}{9}$

$(c)\, Here\, P(|X - 10| \geq c) \leq 0.04$

$\Rightarrow 1 - P(|X - 10| < c) \leq 0.04$

$\Rightarrow 1 - 0.04 \leq P(|X - 10| < c)$

$\Rightarrow 0.96 \leq P(|X - 10| < c)$

$\Rightarrow P(|X - 10| < c) \geq 0.96 - - - - - - - (1)$

Now using the Chebyshev's Inequality

$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2} - - - - - - - (2)$

Comparing equation (1) *and equation* (2) *we get*

$k\sigma = c - - - - - (3)$

$1 - \frac{1}{k^2} = 0.96$

$1 - 0.96 = \frac{1}{k^2}$

$0.04 = \frac{1}{k^2}$

$\Rightarrow k^2 = \frac{1}{0.04} = 25$

$\Rightarrow k = 5$

Put the values of k *and* σ *in equation* (3)*, we get*

$(5)(2) = c$

$\Rightarrow c = 10$

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LAW OF LARGE NUMBERS

Law of large numbers play very important role in the fields of Statistical Modelling and Data Analysis. It tells us relationship about Parameter and Statistic.

STATEMENT: Let $X_1, X_2, \ldots, X_n$ be independent and identically distributed random variables, with $E(X_i)=\mu$ *and* $V(X_i)=\sigma^2<\infty$, $i=1, 2, \ldots, n$. *Then, for any* $\varepsilon>0$

$$\lim_{n\to\infty} P\left(\left|\frac{X_1+X_2+\ldots+X_n}{n}-\mu\right|\geq\varepsilon\right)=0.$$

$$or\ \lim_{n\to\infty} P\left(\left|\overline{X}-\mu\right|>\varepsilon\right)=0.$$

$$or\ \lim_{n\to\infty} P\left(\left|\overline{X}-\mu\right|<\varepsilon\right)=1.$$

PROOF To prove Law of Large Numbers, we shall first find $E(\overline{X})$ *and* $V\left(\overline{X}\right)$

$$E(\overline{X})=E\left(\frac{X_1+X_2+\ldots+X_n}{n}\right)=\frac{1}{n}\left[E(X_1)+E(X_2)+\ldots+E(X_n)\right]=\frac{1}{n}\times n\mu=\mu$$

$$V(\overline{X})=V\left(\frac{X_1+X_2+\ldots+X_n}{n}\right)=\frac{1}{n^2}\left[V(X_1)+V(X_2)+\ldots+V(X_n)\right]=\frac{1}{n^2}\times n\sigma^2=\frac{\sigma^2}{n}$$

Now Chebyshev's Inequality having $E(X)=\mu$ *and* $V(X)=\sigma^2$ is given by

$$P\left(\left|X-\mu\right|<k\sigma\right)\geq 1-\frac{1}{k^2}\ or\ \frac{1}{k^2}\geq 1-P\left(\left|X-\mu\right|<k\sigma\right)\ or\ \frac{1}{k^2}\geq P\left(\left|X-\mu\right|\geq k\sigma\right)$$

$$So\ P\left(\left|X-\mu\right|\geq k\sigma\right)\leq 1/k^2$$

In context to our theorem X is replaced by $\overline{X}$ *and* σ *is replaced by* σ^2/n, So

$$So\ P\left(\left|\overline{X}-\mu\right|\geq k\frac{\sigma}{\sqrt{n}}\right)\leq\frac{1}{k^2}\ this\ holds\ for\ each\ k.So\ let's\ choose\ k=\frac{\varepsilon}{\sigma}\sqrt{n}$$

$$Then\ P\left(\left|\overline{X}-\mu\right|\geq\frac{\varepsilon}{\sigma}\sqrt{n}\times\frac{\sigma}{\sqrt{n}}\right)\leq\frac{1}{(\frac{\varepsilon}{\sigma}\sqrt{n})^2}=\frac{\sigma^2}{\varepsilon^2 n}$$

$$\Rightarrow\quad P\left(\left|\overline{X}-\mu\right|\geq\varepsilon\right)\leq\frac{\sigma^2}{\varepsilon^2 n}$$

$$\Rightarrow\quad \lim_{n\to\infty} P\left(\left|\overline{X}-\mu\right|\geq\varepsilon\right)\leq\lim_{n\to\infty}\frac{\sigma^2}{\varepsilon^2 n}$$

$$\Rightarrow\quad \lim_{n\to\infty} P\left(\left|\overline{X}-\mu\right|\geq\varepsilon\right)=0$$

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CENTRAL LIMIT THEOREM

The central limit theorem (CLT) is one of the most remarkable results in probability theory. The CLT was originally stated and proven by the French Mathematician Pierce-Simon and Marquis de Laplace. *Loosely put, it states that the sum of a large number of independent random variables has a distribution that is approximately normal.* So CLT gives us two important informations

1. It provides a simple method for computing approximate probabilities for sums of independent random variables.
2. It explains the fact that the empirical frequencies of so many natural populations exhibit bell-shaped distribution i.e. Normal Distribution.

We shall discuss CLT in two different cases.

1. CLT for Sum of random variables.
2. CLT for average of random variables.

Let $X_1, X_2, \ldots, X_n$ be a sequence of independent and identically distributed random variables, each having with $mean = \mu$ *and* $\text{var}iance = \sigma^2$. Then

1. If $S = X_1 + X_2 + \ldots + X_n$ *then* $S \sim N(n\mu, n\sigma^2)$

2. If $\overline{X} = \dfrac{X_1 + X_2 + \ldots + X_n}{n}$ *then* $\overline{X} \sim N\left(\mu, \dfrac{\sigma^2}{n}\right)$

STATEMENT OF CENTRAL LIMIT THEOREM

Let $X_1, X_2, \ldots, X_n$ *be a sequence of independent and identically distributed random variables, with* $E(X_i) = \mu$ *and* $V(X_i) = \sigma^2 < \infty$, $i = 1, 2, \ldots, n$. *Then the distribution of*

$$Z_n = \frac{X_1 + X_2 + \cdots + X_n - n\mu}{\sigma\sqrt{n}}$$

converges to the distribution of a standard normal random variable. That is,

$$\lim_{n\to\infty} P(Z_n \le x) = \lim_{n\to\infty} P\left(\frac{X_1 + X_2 + \cdots + X_n - n\mu}{\sigma\sqrt{n}} \le x\right)$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-y^2/2}\, dy.$$

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PROOF OF CENTRAL LIMIT THEOREM

The proof to the central limit theorem given here is a standard one based on the following result of Lévy, which we state without proof.

Theorem 11.13 (Lévy Continuity Theorem) *Let $X_1, X_2, \ldots$ be a sequence of random variables with distribution functions $F_1, F_2, \ldots$ and moment-generating functions $M_{X_1}(t), M_{X_2}(t), \ldots$, respectively. Let X be a random variable with distribution function F and moment-generating function $M_X(t)$. If for all values of t, $M_{X_n}(t)$ converges to $M_X(t)$, then at the points of continuity of F, F_n converges to F.*

Proof of Theorem 11.12 (Central Limit Theorem): Let $Y_n = X_n - \mu$; then $E(Y_n) = 0$ and $\text{Var}(Y_n) = \sigma^2$. We want to prove that the sequence of distribution functions of the random variables $Z_n = (Y_1 + Y_2 + \cdots + Y_n)/(\sigma\sqrt{n})$, $n \geq 1$, converges to the distribution of Z. Since $Y_1, Y_2, \ldots$ are identically distributed, they have the same moment-generating function M. By independence of $Y_1, Y_2, \ldots, Y_n$,

$$\begin{aligned} M_{Z_n}(t) &= E\Big[\exp\Big(\frac{Y_1 + Y_2 + \cdots + Y_n}{\sigma\sqrt{n}}\, t\Big)\Big] = M_{Y_1+Y_2+\cdots+Y_n}\Big(\frac{t}{\sigma\sqrt{n}}\Big) \\ &= M_{Y_1}\Big(\frac{t}{\sigma\sqrt{n}}\Big) M_{Y_2}\Big(\frac{t}{\sigma\sqrt{n}}\Big) \cdots M_{Y_n}\Big(\frac{t}{\sigma\sqrt{n}}\Big) \\ &= \Big[M\Big(\frac{t}{\sigma\sqrt{n}}\Big)\Big]^n. \end{aligned} \tag{11.8}$$

By the Lévy continuity theorem, it suffices to prove that $M_{Z_n}(t)$ converges to $\exp(t^2/2)$, the moment-generating function of Z. Equivalently, it is enough to show that

$$\lim_{n\to\infty} \ln M_{Z_n}(t) = \frac{t^2}{2}. \tag{11.9}$$

Let $h = t/(\sigma\sqrt{n})$; then $n = t^2/(\sigma^2 h^2)$. Thus (11.8) gives

$$\ln M_{Z_n}(t) = n \ln M(h) = \frac{t^2}{\sigma^2 h^2} \ln M(h) = \frac{t^2}{\sigma^2}\Big[\frac{\ln M(h)}{h^2}\Big].$$

Therefore,

$$\lim_{n\to\infty} \ln M_{Z_n}(t) = \frac{t^2}{\sigma^2} \lim_{h\to 0}\Big[\frac{\ln M(h)}{h^2}\Big]. \tag{11.10}$$

Since $M(0) = 1$, $\lim_{h\to 0}\big[\ln M(h)/h^2\big]$ is indeterminate. To determine its value, we apply l'Hôpital's rule twice. We get

$$\begin{aligned} \lim_{h\to 0}\Big[\frac{\ln M(h)}{h^2}\Big] &= \lim_{h\to 0} \frac{M'(h)/M(h)}{2h} = \lim_{h\to 0} \frac{M'(h)}{2hM(h)} \\ &= \lim_{h\to 0} \frac{M''(h)}{2M(h) + 2hM'(h)} = \frac{M''(0)}{2M(0)} = \frac{\sigma^2}{2}, \end{aligned}$$

where $M''(0) = \sigma^2$ since $E(Y_i) = 0$. Thus, by (11.10),

$$\lim_{n\to\infty} \ln M_{Z_n}(t) = \frac{t^2}{\sigma^2}\frac{\sigma^2}{2} = \frac{t^2}{2}.$$

This establishes (11.9) and proves the theorem. ♦

EXAMPLES OF CENTRAL LIMIT THEOREM

EXAMPLE-1 Suppose X denote a random draw from **Binomial Distribution** with mean 15 and standard deviation 3. Let $X_1, X_2, \ldots, X_{50}$ denote 50 draws from the distribution. Let $S = X_1 + X_2 + \ldots + X_{50}$ then

(a) Calculate the mean and variance of S.

(b) Calculate $P(S > 800)$.

SOLUTION *Here* $X \sim$ *Binomial Distribution with mean* $\mu = 15$ & $\sigma^2 = 9$..

Now $S = X_1 + X_2 + \ldots + X_{50}$

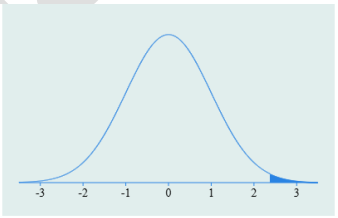
(a) Using Central Limit Theorem *If* $S = X_1 + X_2 + \ldots + X_n$ *then* $S \sim N(n\mu, n\sigma^2)$

Here $S \sim N((50)(15), (50)(9)) \quad \Rightarrow S \sim N(750, 450)$

So mean of S is 750 and variance of S is 450.

(b) $P(S > 800) = ?$ *where* $S \sim N(750, 450)$

Using Continuity Correction $P(S > 800) = P(S > 800.5)$



$$Now\ Z = \frac{S-\mu}{\sigma} = \frac{800.5-750}{\sqrt{450}} = 2.38$$

$$Now\ P(Z > 2.38) = 1 - \Phi(2.38) = 1 - 0.9913 = 0.0087$$

EXAMPLE-2 Let $X_1, X_2, \ldots, X_{20}$ be independent **Poisson random variables** with mean equals 1. Let $B = \frac{X_1 + X_2 + \ldots + X_{20}}{n}$ then

(a) Calculate the mean and variance of B.

(b) Calculate $P(B < 0.93)$.

SOLUTION *Here* $X \sim$ *Poisson Distribution with mean* $\mu = 1$ & $\sigma^2 = 1$..

Now $B = \frac{X_1 + X_2 + \ldots + X_{20}}{n}$

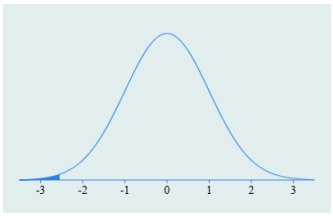
(a) Using Central Limit Theorem *If* $B = \frac{X_1 + X_2 + \ldots + X_n}{n}$ *then* $B \sim N(\mu, \sigma^2/n)$

Here $B \sim N((1), 1/20) \quad \Rightarrow B \sim N(1, 0.05)$

So mean of S is 1 and variance of S is 0.05.

(b) $P(B < 0.93) = ?$ *where* $B \sim N(1, 0.05)$

Using Continuity Correction $P(B < 0.93) = P(B < 0.43)$



$$Now\ Z = \frac{S-\mu}{\sigma} = \frac{0.43-1}{\sqrt{0.05}} = -2.55$$

$$Now\ P(Z < -2.55) = \Phi(-2.55) = 0.0053$$

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EXAMPLE-3 Let $X_1, X_2, \ldots, X_{10}$ be independent random variables, each **uniformly distributed** over $(0,1)$. Let $S = X_1 + X_2 + \ldots + X_{10}$ then calculate $P(S > 6)$.

SOLUTION *Here* $X \sim$ *Uniform Distribution with* $a = 0$ *and* $b = 1$.

Here mean $\mu = \dfrac{a+b}{2} = \dfrac{0+1}{2} = 0.5$ & $\sigma^2 = \dfrac{(b-a)^2}{12} = \dfrac{(1-0)^2}{12} = \dfrac{1}{12}$.

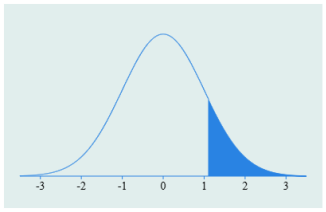
Now $S = X_1 + X_2 + \ldots + X_{10}$

Using Central Limit Theorem *If* $S = X_1 + X_2 + \ldots + X_n$ *then* $S \sim N(n\mu, n\sigma^2)$

Here $S \sim N((10)(0.5), (10)(1/12)) \quad \Rightarrow S \sim N(5, 5/6)$

So mean of S is 5 and variance of S is 5/6.

Now $P(S > 6) = ?$ *where* $S \sim N(5, 5/6)$



Now $Z = \dfrac{6-5}{\sqrt{5/6}} = \dfrac{1}{\sqrt{5/6}} = 1.10$

Now $P(Z > 1.10) = 1 - \Phi(1.10) = 1 - 0.8643 = 0.1357$

EXAMPLE-4 The lifetime of a TV tube (in years) is an **exponential random variable** with mean 10. What is the probability that the average lifetime of a random sample of 36 TV tubes is at least 10.5?

SOLUTION Let X denote the lifetime of a tube

Here $X \sim$ *Exponential Distribution with mean* $= 10 \Rightarrow \lambda = 1/10$.

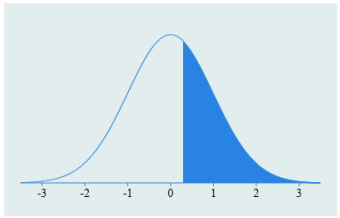
Here mean $\mu = \dfrac{1}{1/10} = 10$ & $\sigma^2 = \dfrac{1}{\lambda^2} = \dfrac{1}{(1/10)^2} = 100$

We have to find $P(\overline{X} \geq 10.5) = ?$

Using Central Limit Theorem *If* $\overline{X} = \dfrac{X_1 + X_2 + \ldots + X_n}{n}$ *then* $\overline{X} \sim N(\mu, \sigma^2/n)$

Here $\overline{X} \sim N\left(10, \dfrac{100}{36}\right) \quad \Rightarrow \overline{X} \sim N(10, 2.78)$

Now $P(\overline{X} \geq 10.5) = ?$ *where* $\overline{X} \sim N(10, 2.78)$



Now $Z = \dfrac{10.5 - 10}{\sqrt{2.78}} = 0.30$

Now $P(Z \geq 0.30) = 1 - \Phi(0.30) = 1 - 0.6179 = 0.3821$

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EXAMPLE-5 The time it takes for a student to finish an aptitude test (in hours) has the probability density function

$$f(x) = \begin{cases} 6(x-1)(2-x) & \text{if } 1 < x < 2 \\ 0 & \text{otherwise.} \end{cases}$$

Approximate the probability that the average length of time it takes for a random sample of 15 students to complete the test is less than 1 hour and 25 minutes.

SOLUTION Let X be the time that it will take for the student in the sample to complete the test.

Here $X \sim$ *Specific Distribution with given p.d.f. We shall calculate its mean &* var*iance.*

$$E(X_i) = \int_1^2 6x(x-1)(2-x)\,dx = 3/2 = 1.5,$$

$$E(X_i^2) = \int_1^2 6x^2(x-1)(2-x)\,dx = 23/10 = 2.3,$$

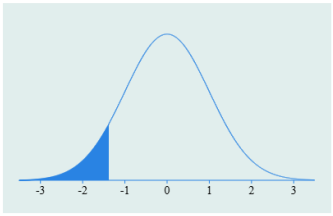
So here mean $\mu = E(X) = 1.5$ & $\sigma^2 = V(X) = E(X^2) - (E(X))^2 = 0.05$

We have to find $P(\overline{X} < 1.42) = ?$ *Note that* 1 *hour* 25 min *utes* $= 1.42$ *hours*

Using Central Limit Theorem *If* $\overline{X} = \dfrac{X_1 + X_2 + \ldots + X_n}{n}$ *then* $\overline{X} \sim N(\mu, \sigma^2/n)$

Here $\overline{X} \sim N\left(1.5, \dfrac{0.05}{15}\right) \quad \Rightarrow \overline{X} \sim N(1.5, 0.0033)$

Now $P(\overline{X} < 1.42) = ?$ *where* $\overline{X} \sim N(1.5, 0.0033)$



Now $Z = \dfrac{1.42 - 1.5}{\sqrt{0.0033}} = -1.39$

Now $P(Z < -1.39) = \Phi(-1.39) = 0.0823$

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